

Force plot: The force plot visualises how different features (input variables) ‘push’ a model’s prediction away from the base value (the average model prediction across the training data).

- Base value (grey line): The average model output across all patients. In this case, it’s around -3.73 .
- $f(x)$ (black text): The final model prediction for this individual = -4.63 (This means the model predicts low sepsis risk).
- Red sections (left push): Features that increase the predicted value (push toward higher sepsis risk).
resp_rate = 24.91 $\rightarrow$ +4.2 SHAP value $\rightarrow$ strongly pushed toward higher risk.
- Blue sections (right push): Features that decrease the predicted value (push toward lower risk).
wbc = 1.03 $\rightarrow$ -3.3 SHAP value
heart_rate = 98.15 $\rightarrow$ -1.56 SHAP value
What this means clinically is that:
- High resp rate (red dots on the right) pushes risk up.
- Low WBC (blue dots on the left) pushes risk down.
- The patient had a high respiration rate, which is typically a red flag for sepsis.
- However, very low white blood cell (WBC) count and moderate heart rate provided strong counter-evidence.
- Net effect: Despite a major risk signal (resp_rate), the stronger negative signals led the model to predict low risk of sepsis.

The force plot gives an interactive summary of how individual features push the prediction away from the base value. Here, features like wbc and heart_rate pulled the model away from predicting sepsis, despite resp_rate pushing in the other direction.

So the final impression is:

- Only one feature (resp_rate) increased the predicted risk,
- But strong negative evidence (low WBC) led the model to conclude low likelihood of sepsis.

The emergence of agentic AI, with the capability to pursue goals, interact with environments, and learn through experience, is a significant revolution in artificial intelligence. Yet, the sophistication of such systems generates trust, accountability, safety, and ethical problems. Explainable AI (XAI) helps overcome these challenges by making the decision-making process of agents transparent, ensuring correct behaviour, determining errors, and promoting trust and cooperation.

The particularity of agentic systems—such

Table 4: How XAI helped explain the result

XAI criteria	What SHAP output shows
Transparency	Doctor can see exactly what raised/lowered risk
Actionability	WBC is very low — probably not sepsis
Trust	Explains decision logic — clinicians more likely to trust model
Auditable	Can show regulators why a patient was (or wasn’t) flagged
Bias detection	Beeswarm shows if age, etc, is unfairly weighted

as dynamic behaviour, goal-orientedness, and learning—requires XAI techniques tailored to such systems. Traditional XAI methods are evolving to accommodate these intricacies, such as enhanced temporal action explanation, goal-oriented behaviour explanation, and visualisation of internal states. Increased adoption of XAI in real-world applications, ranging from autonomous cars to healthcare and industry, underscores its significance. Yet, more research is necessary to enhance XAI methods, particularly in managing complex behaviours and adapting explanations to various users.

Ultimately, whether agentic AI will succeed or not rests with our capacity to make these systems explainable. By making explainability a priority, we can make these agents trustworthy, ethical, and aligned with human values, setting the stage for their use in society. **END** 🐧



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